

Courtney Green

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EDUCATION

Georgetown University, Graduate School of Arts & Sciences, Washington, D.C.

May 2026 (Expected)

Master of Science in Data Science and Analytics

- Relevant Coursework: Probabilistic Modeling and Statistical Computing, Database Systems and SQL, Advanced Data Visualization, Neural Nets & Advanced Deep Learning, Statistical Learning for Analysis, Machine Learning for App Development

University of Virginia, Frank Batten School of Leadership and Public Policy, Charlottesville, VA

May 2024

Bachelor of Arts in Public Policy and Leadership; Minor in Computer Science

TECHNICAL SKILLS

- **Programming Languages:** R, Python, SQL, Java, JavaScript, HTML, C, C#, C++
- **Software:** AWS, GitHub, Tableau, Excel, Stata, Jupyter Notebook, Docker, Visual Studio Code, Airtable, WordPress

RELEVANT EXPERIENCE

EVgo, Los Angeles, CA

June 2024 – August 2024

Market Development & Public Policy Intern

- Researched and analyzed government incentives, including the 30C tax credit and NEVI program funds, to support the development of electric vehicle charging infrastructure.
- Drafted and optimized data-driven reports, evaluating electricity rate designs and charging infrastructure development.
- Collaborated on cross-functional teams, working closely with software developers, policy experts, and data analysis teams to support infrastructure expansion through the Department of Energy's Title 17 Clean Financing program.

VOX Global, Washington D.C.

May 2023 – August 2023

Public Affairs Intern

- Performed comprehensive audits and data collection to inform client reports, using analytical tools like Excel and Tableau to identify trends in corporate social responsibility (CSR) initiatives.
- Developed media outreach strategies with team members, managing stakeholder engagement data, and supporting decision-making for Fortune 500 clients.
- Wrote, edited, and proofread various reports and internal communications, helping streamline client information-sharing processes.

Mindset, Washington D.C.

June 2021 – August 2022

Policy Research Intern

- Performed qualitative and quantitative research on cybersecurity and privacy, analyzing policy frameworks and conducting risk assessments to provide actionable recommendations for clients.
- Tracked legislative trends in cybersecurity and privacy to assess the potential impacts of regulatory changes for clients.
- Created and regularly maintained an extensive cybersecurity and privacy legislative database of congressional floor activity, the Biden Administration, and independent regulators for clients.

PROJECTS

Probabilistic Modeling of Prosecutorial Misconduct and Over-Policing in Chicago

Fall 2024

Technology Stack: Python (Pandas, Scikit-learn, GeoPandas, Plotly), Quarto, Git, PCA, t-SNE, DBSCAN

- Cleaned and merged exoneration, arrest, incarceration, and misconduct datasets; engineered a counterfactually balanced dataset for supervised modeling.
- Trained logistic regression and random forest classifiers to predict wrongful conviction risk (ROC-AUC: 0.95); applied PCA and t-SNE for dimensionality reduction.
- Used K-Means and DBSCAN to identify hidden spatial and demographic patterns; built a reproducible Quarto site to visualize geospatial and clustering results.

Predicting Religious Affiliation from Open-Ended Survey Responses Using Neural Networks

Spring 2025

Technology Stack: Python (TensorFlow, Keras, Scikit-learn), NLTK, Matplotlib, Git

- Preprocessed 12,000+ multilingual open-ended survey responses using tokenization, lemmatization, and TF-IDF vectorization.
- Trained deep feedforward neural networks with dropout regularization for multi-class classification; benchmarked against logistic regression, Support Vector Machine (SVM), and random forest.
- Evaluated models using cross-validation, confusion matrices, and F1 scores; visualized training dynamics to inform tuning.

Modeling Voter Turnout and Partisan Lean in the 2024 U.S. Presidential Election

Spring 2025

Technology Stack: Python (Scikit-learn, Pandas, Matplotlib), R, Git, Logistic Regression, SVM, KNN, Random Forest

- Cleaned and merged county-level demographic and election data; engineered features to classify partisan lean (Democrat-leaning vs. non-Democrat-leaning) and voter turnout (high vs. low).
- Trained and compared classification models (Logistic Regression, SVM, KNN, Random Forest) using best subset selection, cross-validation, and validation curves.
- Simulated election-night reporting dynamics using early vs. late-reporting states; evaluated model performance using accuracy, precision, recall, and ROC-AUC.